

## Practical No: 09

Write a Program for Training and use word embedding Word2vec/GloVe.

```
In [4]: !pip install --upgrade pip
!pip install numpy==1.23.5
!pip install gensim==4.3.2
!pip install smart_open
```

```

Requirement already satisfied: pip in /usr/local/lib/python3.11/dist-packages (2
4.1.2)
Collecting pip
  Downloading pip-25.0.1-py3-none-any.whl.metadata (3.7 kB)
Downloading pip-25.0.1-py3-none-any.whl (1.8 MB)
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Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.1.2
    Uninstalling pip-24.1.2:
      Successfully uninstalled pip-24.1.2
Successfully installed pip-25.0.1
Collecting numpy==1.23.5
  Downloading numpy-1.23.5-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_6
4.whl.metadata (2.3 kB)
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hl (17.1 MB)
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Installing collected packages: numpy
  Attempting uninstall: numpy
    Found existing installation: numpy 1.26.4
    Uninstalling numpy-1.26.4:
      Successfully uninstalled numpy-1.26.4
ERROR: pip's dependency resolver does not currently take into account all the pac
kages that are installed. This behaviour is the source of the following dependenc
y conflicts.
thinc 8.3.6 requires numpy<3.0.0,>=2.0.0, but you have numpy 1.23.5 which is inco
mpatible.
chex 0.1.89 requires numpy>=1.24.1, but you have numpy 1.23.5 which is incompatib
le.
xarray 2025.1.2 requires numpy>=1.24, but you have numpy 1.23.5 which is incompat
ible.
bigframes 1.42.0 requires numpy>=1.24.0, but you have numpy 1.23.5 which is incom
patible.
alumentations 2.0.5 requires numpy>=1.24.4, but you have numpy 1.23.5 which is i
ncompatible.
scikit-image 0.25.2 requires numpy>=1.24, but you have numpy 1.23.5 which is inco
mpatible.
jax 0.5.2 requires numpy>=1.25, but you have numpy 1.23.5 which is incompatible.
albucore 0.0.23 requires numpy>=1.24.4, but you have numpy 1.23.5 which is incomp
atible.
blosc2 3.3.0 requires numpy>=1.26, but you have numpy 1.23.5 which is incompatibl
e.
jaxlib 0.5.1 requires numpy>=1.25, but you have numpy 1.23.5 which is incompatibl
e.
treescope 0.1.9 requires numpy>=1.25.2, but you have numpy 1.23.5 which is incomp
atible.
imbalanced-learn 0.13.0 requires numpy<3,>=1.24.3, but you have numpy 1.23.5 whic
h is incompatible.
pymc 5.21.2 requires numpy>=1.25.0, but you have numpy 1.23.5 which is incompatib
le.
tensorflow 2.18.0 requires numpy<2.1.0,>=1.26.0, but you have numpy 1.23.5 which
is incompatible.
Successfully installed numpy-1.23.5

```

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Collecting gensim==4.3.2
  Downloading gensim-4.3.2-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (8.3 kB)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.11/dist-packages (from gensim==4.3.2) (1.23.5)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.11/dist-packages (from gensim==4.3.2) (1.13.1)
Requirement already satisfied: smart-open>=1.8.1 in /usr/local/lib/python3.11/dist-packages (from gensim==4.3.2) (7.1.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.11/dist-packages (from smart-open>=1.8.1->gensim==4.3.2) (1.17.2)
Downloading gensim-4.3.2-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (26.7 MB)
_____ 26.7/26.7 MB 142.4 MB/s eta 0:00:00
ERROR: Operation cancelled by user
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```

In [2]: !pip install gensim

```

Collecting gensim
  Downloading gensim-4.3.3-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (8.1 kB)
Collecting numpy<2.0,>=1.18.5 (from gensim)
  Downloading numpy-1.26.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)
_____ 61.0/61.0 kB 4.4 MB/s eta 0:00:00
Collecting scipy<1.14.0,>=1.7.0 (from gensim)
  Downloading scipy-1.13.1-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (60 kB)
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Requirement already satisfied: smart-open>=1.8.1 in /usr/local/lib/python3.11/dist-packages (from gensim) (7.1.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.11/dist-packages (from smart-open>=1.8.1->gensim) (1.17.2)
Downloading gensim-4.3.3-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (26.7 MB)
_____ 26.7/26.7 MB 46.2 MB/s eta 0:00:00
Downloading numpy-1.26.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (18.3 MB)
_____ 18.3/18.3 MB 57.7 MB/s eta 0:00:00
Downloading scipy-1.13.1-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (38.6 MB)
_____ 38.6/38.6 MB 14.2 MB/s eta 0:00:00
Installing collected packages: numpy, scipy, gensim
  Attempting uninstall: numpy
    Found existing installation: numpy 2.0.2
    Uninstalling numpy-2.0.2:
      Successfully uninstalled numpy-2.0.2
  Attempting uninstall: scipy
    Found existing installation: scipy 1.14.1
    Uninstalling scipy-1.14.1:
      Successfully uninstalled scipy-1.14.1
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
thinc 8.3.6 requires numpy<3.0.0,>=2.0.0, but you have numpy 1.26.4 which is incompatible.
Successfully installed gensim-4.3.3 numpy-1.26.4 scipy-1.13.1

```

```
In [1]: import gensim
        from gensim import corpora
```

```
In [4]: from gensim.models import Word2Vec
        from nltk.tokenize import word_tokenize
        import nltk

        # Download the punkttokenizer models from NLTK
        nltk.download('punkt')
        nltk.download('punkt_tab')

        # Function to train Word2Vec model
        def train_word_embeddings(sentences):
            # Tokenize sentences using NLTK word_tokenize and convert to lowercase
            tokenized_sentences = [word_tokenize(sentence.lower()) for sentence in sentences]

            # Train Word2Vec model
            model = Word2Vec(sentences=tokenized_sentences, vector_size=100, window=5, min_count=1)

            return model

        # Function to use trained Word2Vec model and find similar words
        def use_word_embeddings(model, word, top_n=5):
            try:
                # Get the top N similar words to the input word
                similar_words = model.wv.most_similar(word, topn=top_n)
                print(f"Words most similar to '{word}':")
                for w, score in similar_words:
                    print(f"{w}: {score:.4f}")
            except KeyError:
                print(f"'{word}' not in vocabulary")

        # Example usage
        sentences = [
            "The quick brown fox jumps over the lazy dog",
            "A fox is a cunning animal",
            "The dog barks at night",
            "Foxes and dogs are different species"
        ]

        # Train Word2Vec model using the provided sentences
        model = train_word_embeddings(sentences)

        # Use the trained model to find words similar to "fox"
        use_word_embeddings(model, "fox")
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
```

```
Words most similar to 'fox':
at: 0.1607
dogs: 0.1593
barks: 0.1372
night: 0.1230
and: 0.0854
```